

# PHISHING EMAIL DETECTION & AWARENESS REPORT

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Task 2 — Cybersecurity Internship Program

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| Prepared By:     | Nancy Cafti                      |
| Date:            | 7 <sup>th</sup> May 2026         |
| Dataset:         | Real-World Online Phishing Cases |
| Emails Analyzed: | 5 (Mix: Phishing & Suspicious)   |
| Submission:      | 8 <sup>th</sup> May 2026         |

## 1. Executive Summary

Phishing is the number one cyber threat facing individuals and businesses today. According to Proofpoint's 2024 State of the Phish report, 86% of organizations experienced at least one phishing attempt in 2024, and the UK government's Cyber Security Breaches Survey 2025 found that phishing accounted for 93% of all cyber crime. IBM's Cost of a Data Breach Report 2024 puts the average cost of a phishing-related breach at \$5.1 million.

This report analyzes 5 real phishing email cases documented from trusted cybersecurity sources. Each email impersonates a well-known brand (PayPal, Microsoft, Apple, Netflix, and the IRS) and uses social engineering to trick ordinary people into handing over their credentials, financial details, or money.

The report classifies each email, explains the phishing techniques used in plain, non-technical language, and provides actionable prevention guidelines for users at all levels.

| Metric                           | Value                                  |
|----------------------------------|----------------------------------------|
| Total emails analyzed            | 5                                      |
| Classified as Phishing           | 4 emails                               |
| Classified as Suspicious         | 1 email                                |
| Classified as Safe               | 0 emails                               |
| Brands impersonated              | PayPal, Microsoft, Apple, Netflix, IRS |
| Primary attack goal              | Credential theft & financial fraud     |
| Average reported cost (IBM 2024) | \$5.1 million per breach               |

## 2. Methodology

This analysis follows a structured, repeatable approach used by cybersecurity professionals to evaluate suspicious emails:

1. Source Collection — Five real phishing email cases were identified and collected from reputable cybersecurity sources including Norton, NordVPN, Keepnet Labs, Certi-ID, and MailGuard (all 2024–2025 documented cases).
2. Sender Analysis — The sender display name, actual email address, and domain were examined to detect spoofing or impersonation.
3. Link Inspection — All embedded links and URLs in each email were noted and analyzed for mismatch, typo-squatting, or redirect to fake login pages.
4. Content Analysis — The email body was reviewed for urgency language, threats, unrealistic promises, grammar issues, and generic greetings.
5. Classification — Each email was assigned one of three labels: PHISHING, SUSPICIOUS, or SAFE, based on the number and severity of indicators found.
6. Reporting — All findings were documented in plain, business-friendly language accessible to non-technical readers.

Tools Referenced: MX=Toolbox Email Header Analyzer, Virus Total URL scanner, browser developer tools for link inspection.

### 3. Email Sample Analysis

The following five emails were collected from documented real-world phishing cases. Each email impersonates a trusted brand to steal sensitive information from victims.

#### Email 1 — Fake PayPal Suspicious Activity Alert

|                       |                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------|
| Sender Name / Display | PayPal Support <service@paypa1-secure.com>                                                    |
| Sender Domain         | paypa1-secure.com (NOT paypal.com — '1' replaces 'l')                                         |
| Subject Line          | Action Required: Suspicious Activity Detected on Your Account                                 |
| Malicious Link / URL  | <a href="http://paypa1-secure.com/verify-account">http://paypa1-secure.com/verify-account</a> |
| Classification        | <b>PHISHING</b>                                                                               |

#### What This Email Claims:

The email pretends to be from PayPal and warns the recipient that 'suspicious activity' has been detected on their account. It urges them to click a link immediately and log in to 'verify' their account — or risk having it permanently suspended.

#### Red Flags Identified:

| Phishing Indicator | What Was Found                                                                                                                  | Risk Level      |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Spoofed Domain     | paypa1-secure.com uses the number '1' instead of the letter 'l' — a classic typo-squatting trick designed to fool quick readers | <b>CRITICAL</b> |
| Generic Greeting   | Says 'Dear Customer' instead of your actual name — PayPal always addresses you by your registered first name                    | <b>HIGH</b>     |
| Fake Urgency       | Threatens account suspension to pressure you into acting immediately without thinking                                           | <b>HIGH</b>     |
| Malicious Link     | The link leads to a fake PayPal login page that captures your username and password                                             | <b>CRITICAL</b> |
| Unsolicited Alert  | You never requested a security review — legitimate PayPal alerts follow a specific pattern and link to paypal.com only          | <b>MEDIUM</b>   |

#### In Plain English:

This email is a trap. The fake domain looks almost exactly like the real PayPal address. If you click the link and enter your login details, a criminal receives them instantly. They can then drain your PayPal balance or make purchases using your linked cards. Always go to paypal.com directly by typing it yourself — never through a link in an email.

Source: Keepnet Labs — 30 Phishing Email Examples (2024); NordVPN Phishing Examples (2026)

Email 2 — Fake Microsoft Teams Login Alert

|                       |                                                                 |
|-----------------------|-----------------------------------------------------------------|
| Sender Name / Display | Microsoft Teams <noreply@micros0ft-teams.net>                   |
| Sender Domain         | micros0ft-teams.net (zero '0' replaces letter 'o' in Microsoft) |
| Subject Line          | You Have a New Message Waiting — Sign In to View                |
| Malicious Link / URL  | http://micros0ft-teams.net/signin                               |
| Classification        | PHISHING                                                        |

What This Email Claims:

The email impersonates Microsoft Teams and notifies the recipient that a colleague has sent them an urgent message. To read it, they must click a button to sign in. The link takes them to a convincing fake Microsoft Teams login page.

Red Flags Identified:

| Phishing Indicator    | What Was Found                                                                                                                       | Risk Level |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------|
| Typo-squatted Domain  | micros0ft-teams.net replaces the letter 'o' with zero '0' — nearly invisible to the human eye, especially on mobile screens          | CRITICAL   |
| Not microsoft.com     | Legitimate Microsoft Teams notifications always come from microsoft.com or teams.microsoft.com — never a third-party domain          | CRITICAL   |
| Credential Harvesting | The fake login page is designed to look identical to the real Teams login — all entered credentials go directly to the attacker      | CRITICAL   |
| Vague Message Bait    | Does not say who the message is from or what it is about — designed to trigger curiosity without giving you enough detail to verify  | HIGH       |
| Mobile Exploit        | 85% of users open emails on phones where only the display name shows — making the spoofed domain invisible without tapping to expand | HIGH       |

In Plain English:

This scam targets remote workers and office teams who use Microsoft Teams daily. The email looks completely normal at first glance — but the web address has one character swapped. If you enter your Microsoft password on the fake page, attackers gain access to your entire Microsoft account including Outlook, OneDrive, and SharePoint. Always check that login pages show teams.microsoft.com in your browser address bar.

Source: IT Governance — How to Spot a Phishing Email (2025); GRC Solutions Phishing Guide (2025)

Email 3 — Fake Apple ID Account Locked Alert

|                       |                                                                 |
|-----------------------|-----------------------------------------------------------------|
| Sender Name / Display | Apple <no-reply@apple-id-security.support>                      |
| Sender Domain         | apple-id-security.support (NOT apple.com)                       |
| Subject Line          | Your Apple ID Has Been Locked — Immediate Verification Required |
| Malicious Link / URL  | http://apple-id-security.support/unlock                         |
| Classification        | PHISHING                                                        |

What This Email Claims:

The email pretends to be from Apple and claims your Apple ID has been locked due to 'multiple failed sign-in attempts.' It instructs you to click a link and verify your identity by entering your Apple ID, password, and sometimes your credit card details to 'confirm your identity.'

Red Flags Identified:

| Phishing Indicator    | What Was Found                                                                                                                | Risk Level |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|------------|
| Fake Domain           | apple-id-security.support is NOT owned by Apple — Apple only sends emails from @apple.com or @id.apple.com                    | CRITICAL   |
| Fear Tactic           | Claiming your account is 'locked' triggers immediate panic — a deliberate tactic to bypass rational thinking                  | HIGH       |
| Requests Payment Info | Asking for a credit card to 'verify identity' is never something Apple does in an email — this is pure data theft             | CRITICAL   |
| .support TLD          | The use of a .support domain instead of .com is a giveaway — scammers buy cheap alternative domain extensions to mimic brands | HIGH       |
| No Personalization    | Does not reference your actual Apple device, purchase history, or name — a mass-sent phishing blast, not a personal alert     | MEDIUM     |

In Plain English:

Apple will never email you asking for your credit card details to unlock an account. If your Apple ID is ever truly locked, you will be directed to appleid.apple.com — which you should visit by typing it yourself. This email is designed to steal both your Apple account AND your financial information in a single attack.

Source: NordVPN — 15 Common Phishing Email Examples (2026); Norton — 10 Real Phishing Email Examples (2024)

Email 4 — Fake Netflix Payment Failure Notice

|                       |                                                                   |
|-----------------------|-------------------------------------------------------------------|
| Sender Name / Display | Netflix <billing@netflix-payments.info>                           |
| Sender Domain         | netflix-payments.info (NOT netflix.com)                           |
| Subject Line          | Your Netflix Subscription Has Been Suspended — Update Billing Now |
| Malicious Link / URL  | http://netflix-payments.info/update-billing                       |
| Classification        | PHISHING                                                          |

What This Email Claims:

The email impersonates Netflix and claims the recipient's payment method was declined. It warns that their account will be permanently cancelled unless they update their billing information within 24 hours by clicking the provided link.

Red Flags Identified:

| Phishing Indicator  | What Was Found                                                                                                                   | Risk Level |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|------------|
| Fake Billing Domain | netflix-payments.info is not affiliated with Netflix — Netflix only communicates from @netflix.com                               | CRITICAL   |
| Artificial Deadline | 'Within 24 hours' is a pressure tactic designed to prevent you from pausing to verify the email's legitimacy                     | HIGH       |
| Payment Data Theft  | The fake page collects full credit card number, expiry date, and CVV — enough to make fraudulent purchases anywhere              | CRITICAL   |
| .info Domain        | Legitimate subscription services use .com — .info domains are cheap, widely used by scammers, and carry no corporate credibility | HIGH       |
| Generic Salutation  | Real Netflix emails address you by your account name — 'Dear Member' signals a mass phishing blast                               | MEDIUM     |

In Plain English:

Streaming service phishing emails are among the most common because almost everyone has a subscription. If your Netflix payment actually fails, you will see a notification inside the Netflix app or website — not through an urgent external email. Never enter your card details through an email link. Go to netflix.com directly and check your account there.

Source: MailGuard — Netflix Phishing Email Analysis (2021, pattern continues through 2025); Norton Phishing Examples (2024)



Email 5 — Fake IRS Tax Refund Notice (Suspicious)

|                       |                                                              |
|-----------------------|--------------------------------------------------------------|
| Sender Name / Display | Internal Revenue Service <refund@irs-tax-refund.com>         |
| Sender Domain         | irs-tax-refund.com (NOT irs.gov — uses .com instead of .gov) |
| Subject Line          | You Are Eligible for a Tax Refund — Submit Your Details      |
| Malicious Link / URL  | http://irs-tax-refund.com/claim-refund                       |
| Classification        | SUSPICIOUS (High Risk)                                       |

What This Email Claims:

The email poses as the US Internal Revenue Service (IRS) and tells the recipient they are owed a tax refund. To receive the money, they must click a link and submit personal details including their Social Security Number, date of birth, and bank account information.

Red Flags Identified:

| Phishing Indicator     | What Was Found                                                                                                           | Risk Level |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|------------|
| Wrong Domain Extension | The IRS only operates from .gov domains (irs.gov) — any email from a .com address claiming to be the IRS is fraudulent   | CRITICAL   |
| Unsolicited Refund     | The IRS does not initiate contact by email to notify you of refunds — they always send official letters by postal mail   | HIGH       |
| Requests SSN by Email  | Asking for a Social Security Number over email or a website link is a major red flag — this enables full identity theft  | CRITICAL   |
| Too Good to Be True    | A surprise refund you never applied for should always raise suspicion — scammers use reward promises to lower your guard | HIGH       |
| Classified Suspicious  | This email could be a test/awareness email if found in a corporate environment — still treat as dangerous until verified | MEDIUM     |

Why Classified as 'Suspicious' and not 'Phishing':

This email is classified as Suspicious (rather than confirmed Phishing) because in some corporate security awareness training programs, IRS-themed emails are used as simulated tests. However, the real-world version of this email IS a confirmed phishing attack. Any unsolicited email claiming to be from the IRS requesting personal or financial details should be treated as Phishing and deleted immediately.

Source: Norton — IRS Phishing Email Examples (2024); IT Governance — IRS ZIP Malware Campaign March 2025

## 4. Classification Summary

The table below summarizes the classification of all 5 emails analyzed in this report:

| # | Brand Impersonated | Attack Type                                   | Classification                                                                                      | Primary Risk               |
|---|--------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------|
| 1 | PayPal             | Credential Harvesting via Typosquatted Domain |  <b>PHISHING</b>   | Account & Payment Fraud    |
| 2 | Microsoft          | Fake Teams Login — Credential Theft           |  <b>PHISHING</b>   | Corporate Account Breach   |
| 3 | Apple              | Account Lock Scare + Payment Data Theft       |  <b>PHISHING</b>   | Identity & Financial Theft |
| 4 | Netflix            | Billing Failure — Credit Card Harvesting      |  <b>PHISHING</b>   | Credit Card Fraud          |
| 5 | IRS                | Fake Tax Refund — Social Engineering          |  <b>SUSPICIOUS</b> | Identity Theft (SSN)       |

| Label                                                                                                 | What It Means                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>PHISHING</b>    | Confirmed malicious email. Contains spoofed sender, fake domain, and malicious link designed to steal credentials or money. Delete immediately and report. |
|  <b>SUSPICIOUS</b> | High-risk email with multiple warning signs. May be a phishing attempt or a security awareness test. Do NOT click links — verify with your IT team first.  |
|  <b>SAFE</b>       | No phishing indicators detected. Sender domain is legitimate, links match the brand domain, and the content matches expected communication patterns.       |

## 5. Phishing Techniques Explained

All 5 emails in this report used one or more of the following techniques. Here is what each one means in everyday language:

| Technique                    | Plain English Explanation                                                                                                                                                                             | Seen In           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Typosquatting</b>         | Buying a domain name that looks almost identical to a real one — e.g. paypa1.com vs paypal.com. Criminals count on you not noticing the one-character difference, especially on a small phone screen. | Emails 1, 2, 3, 4 |
| <b>Brand Impersonation</b>   | Copying the logo, colors, and writing style of a trusted company like Apple or Microsoft to make the email look 100% real. The look is real — the sender is not.                                      | All 5 emails      |
| <b>Fear &amp; Urgency</b>    | Creating panic with words like 'Your account is locked', 'Act within 24 hours', or 'Permanent suspension.' Fear shuts down critical thinking and makes people click before they consider.             | Emails 1, 3, 4, 5 |
| <b>Credential Harvesting</b> | Linking to a fake login page that looks identical to the real website. When you type your password, it goes to a criminal — not to the real company.                                                  | Emails 1, 2, 3, 4 |
| <b>Social Engineering</b>    | Manipulating human emotions and trust rather than hacking technology. These emails work because people trust PayPal, Apple, and the government — not because the technology is advanced.              | All 5 emails      |
| <b>Too-Good-To-Be-True</b>   | Promising unexpected rewards (tax refunds, free money) to make victims feel lucky and lower their defenses. The promise is fake — the data they collect is real.                                      | Email 5           |

## 6. Prevention & Awareness Guidelines

We do not need to be a cybersecurity expert to protect yourself. We have to follow these rules and we will avoid the vast majority of phishing attacks:

### Rule 1 — Always Check the Actual Email Address

The display name (e.g. 'PayPal Support') can be anything the sender wants it to be. What matters is the real email address behind it. Tap or hover over the sender name to reveal it.

- Legitimate PayPal emails only come from @paypal.com
- Legitimate Microsoft emails only come from @microsoft.com
- The IRS only uses @irs.gov — any .com address claiming to be the IRS is fake

### Rule 2 — Never Click Links in Unexpected Emails

If you receive an unexpected alert about your account, do NOT click the link in the email. Instead, open your browser and type the company's real website address yourself.

- Suspicious PayPal email? Go to paypal.com yourself
- Suspicious Apple email? Go to appleid.apple.com yourself
- This one habit alone blocks nearly all phishing attacks

### Rule 3 — Slow Down on 'Urgent' Emails

Scammers deliberately create panic so you act before you think. Any email that says 'Act Now,' 'Within 24 Hours,' or 'Your Account Will Be Closed' is designed to rush you into a mistake.

- Legitimate companies give you days or weeks to resolve account issues
- Take 2 minutes to verify before you click anything

### Rule 4 — Never Enter Payment Details Through Email Links

No legitimate company — PayPal, Netflix, Apple, your bank — will ever ask you to enter credit card details through a link in an email. This is always a scam.

- Update billing only by logging into the official app or website
- If in doubt, call the company's official customer service number

### Rule 5 — Check the URL Before You Login

Before entering any username or password on a website, look at the address bar in your browser. Make sure the domain is exactly right.

- paypal.com is safe. paypa1-secure.com is a scam
- teams.microsoft.com is safe. micros0ft-teams.net is a scam
- Look for the padlock icon AND the correct domain name

## Rule 6 — Enable Two-Factor Authentication (2FA)

Even if a scammer steals your password, two-factor authentication stops them from logging in. Enable it on all important accounts — email, banking, PayPal, Apple ID, Microsoft.

- Use an authenticator app (Google Authenticator, Microsoft Authenticator) rather than SMS where possible

## Rule 7 — Report Phishing Emails

Do not just delete suspicious emails — reporting them helps protect everyone else too.

- Gmail: Three dots menu > 'Report phishing'
- Outlook: 'Report message' button > 'Phishing'
- Forward to [reportphishing@apwg.org](mailto:reportphishing@apwg.org) (Anti-Phishing Working Group)
- US users: Forward IRS phishing to [phishing@irs.gov](mailto:phishing@irs.gov)

## Rule 8 — When in Doubt, Ask Your IT Team

If you receive an email that feels wrong but you are not sure, do not guess — ask. Forward it to your IT or security team before taking any action. It takes 30 seconds and could prevent a major breach.

## 7. Conclusion

All 5 emails analyzed in this report followed a clear and predictable pattern: impersonate a trusted brand, create a sense of fear or urgency, and trick the victim into clicking a link that steals their credentials or financial information.

None of these attacks required advanced technology. They succeeded because they looked familiar and triggered an emotional response. The most powerful defense is not software — it is an informed, aware user who knows what to look for.

Phishing attacks are increasing in sophistication. AI tools are now helping criminals write more polished, personalized emails that no longer contain the spelling mistakes and awkward grammar of older scams. This makes awareness training more important than ever.

| If you see this...                           | Do this                                                           |
|----------------------------------------------|-------------------------------------------------------------------|
| An urgent email from a well-known brand      | <b>Check the actual email address — not just the display name</b> |
| A link asking you to log in                  | <b>Go to the website yourself — never click the link</b>          |
| A promise of a refund, prize, or inheritance | <b>Delete it — no one gives money to strangers via email</b>      |
| A request for your card or bank details      | <b>Never share these via email — report the email immediately</b> |
| An email threatening account closure         | <b>Slow down — verify by calling the company directly</b>         |
| Something that just feels wrong              | <b>Trust your instincts — report it to your IT team</b>           |

— End of Report —

Submitted via GitHub Public Repository | May 2026 | Task 2 — Phishing Email Detection & Awareness